Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 32)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 compares organic and inorganic fertilisers.

0 1 . 1 Complete Table 1 by giving two advantages and two disadvantages of each type of fertiliser.

[6 marks]

Table 1

Fertiliser	Advantages	Disadvantages
Organic (manure, sewage sludge, compost)		
Inorganic (eg ammonium nitrate, NPK)		

0 1 . 2 Explain how hydroponic cultivation maximises productivity and give **two** disadvantages.

[4 marks]

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0 2 Table 1 shows features of some fishing methods.

0 2 . 1 Complete Table 1.

[6 marks]

Table 1

Method	Target species	One environmental disadvantage	
Demersal trawling			
Purse seining			
Pelagic long-lining			

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0 2 . 2 Explain why by-catch is a particular problem with shrimp trawling and drift netting.

[2 marks]

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0 2 . 3 Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

10

0 3 Table 1 shows the number of bird species and butterfly species recorded in three woodlands in the same region.

Table 1

Woodland	Management	Bird species	Butterfly species
A	Ancient oak wood, coppiced on rotation with clearings	42	23
B	Ancient oak wood, unmanaged for 60 years, dense canopy	28	9
C	40-year-old conifer plantation	14	3

0 3 . 1 Use Table 1 and your knowledge to explain the differences in biodiversity between the three woodlands.

[5 marks]

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0 3 . 2 Describe **three** conservation efforts used to protect UK woodlands.

[3 marks]

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0 3 . 3 Explain why a high biodiversity increases the ecological stability of a woodland.

[2 marks]

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0 4 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 4 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 4 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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0 5 . 1 Explain how natural material cycles differ from linear human systems of material use.

[4 marks]

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0 5 . 2 Explain why the use of toxic metals in electronic products is a particular problem for sustainability.

[2 marks]

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0 5 . 3 Describe how the design of a product can move it from a linear to a cyclical system.

[4 marks]

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0 6 . 1 Describe how soil texture is determined using a stack of sieves.

[3 marks]

0 6 . 2 Explain why a soil auger gives a more representative sample than a trowel.

[2 marks]

0 6 . 3 Explain why the organic matter content will be overestimated if a soil sample is heated above 550 °C.

[2 marks]

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0 6 . 4 Describe how a turbidimeter measures water turbidity and explain how quantitative results are obtained.

[3 marks]

10

0 7 Table 1 shows some designated protected areas in the UK.

0 7 . 1 Complete Table 1 to show the legislation under which each area is designated and its key feature.

[4 marks]

Table 1

Designation	Designated under	Key feature	
Site of Special Scientific Interest (SSSI)			
Special Protection Area (SPA)			
Ramsar site			
Marine Conservation Zone (MCZ)			

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0 7 . 2 Explain how designating a protected area protects habitats and species.

[3 marks]

0 7 . 3 Explain why voluntary agri-environmental schemes are needed in addition to legislation.

[3 marks]

10

0 8 . 1 Explain **four** properties of water that make it essential for life.

[4 marks]

0 8 . 2 Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

[4 marks]

0 8 . 3 Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

[2 marks]

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0 9 . 1 Explain what is meant by **biomimetics** and describe **three** examples.

[4 marks]

0 9 . 2 Explain why many plants produce chemicals that are useful as human medicines, and give **two** examples.

[3 marks]

0 9 . 3 Explain why the discovery of medicines is a strong argument for conserving whole habitats rather than individual species.

[2 marks]

0 9 . 4 Give **one** example of a wildlife species used in physiological research and the human health issue it helps to understand.

[1 mark]

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1 0 . 1 Discuss the strategies that can be used to increase the sustainability of agriculture.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **permaculture**.

[1 mark]

10

Write an essay on one of the following topics.

11 . 1 Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

OR

11 . 2 Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS